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Polka-dotted Stars: Mapping the Surface of HAT-P-11

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ABSTRACT

1. INTRODUCTION

The quest to unravel the mysteries of distant planetary systems has led to a fast-evolving field of research, where exoplanet studies are at the forefront of astronomical investigations. Characterizing the surfaces of exoplanet-host stars is crucial for understanding the environments and evolution of their planetary systems. For example, the Sun's magnetic field drives the star's activity cycles, the heating of its outer atmosphere and the solar wind, which affects the Earth's magnetosphere and biosphere (Babcock 1961; Charbonneau 2014). Therefore, constructing a catalogue of the magnetic properties of other stars similar to the Sun would support the quest towards understanding the population of detected exoplanets and the development of a highly generalizable theory on stellar magnetic dynamo.

The reverse statement is also true. The exisiting popultion of exoplanets along with their orbital parameters can help characterize their host-stars in terms of their surface magnetic activity. Stellar surface features like spots and flares impact observables in planet-detection research including photometric variability, spectral line profiles, and radial velocity shifts. However, directly resolving stellar surfaces remains challenging.

While high-resolution spectroscopy provides the most robust method of determining stellar parameters, obtaining such data for large samples remains observationally expensive. Some recent works offered invaluable insights into the topographical characteristics of stars and their magnetic activity, responsible for the observed variatians on their surfaces. For example, the influence of star spots can be discerned through photometric data acquired from Kepler or TESS. Luger et al. (2021a) investigated the star spots features on the stellar surfaces using Gaussian Process (GPs) with a *physically interpretable* kernel. They showed that it is hard to put constrains on star spot properties if you only have a light curve for one star, it is rather usefule though to perform the same analysis for an ensemble of stars. However, when a star has an exoplanet in orbit, the spot features become observable in the transit data, providing valuable constraints on the properties of these spots (whether they are dark or bright, their sizes, and their latitudes).

Data-driven approaches like Gaussian processes (GPs) offer pathways to map the surfaces of exoplanet hosts through measurable disk-integrated signals. In this work, we develop a Gaussian process framework to model the surfaces of exoplanet-host stars based on time-series photometry. We utilize expressive kernel function from Luger et al. (2021a). We first use this model on a simulated data to test the model, then we apply the model to an active K-dwarf HAT-P-11.

The planet-hosting star HAT-P-11 has been extensively studied for its star spots. HAT-P-11 is orbited by a Neptune-sized exoplanet, HAT-P-11b, which was discovered in 2008 via the transit method (Bakos et al. 2010; Deming et al. 2011). The transiting nature of HAT-P-11b has allowed for detailed characterization of the star spot activity on HAT-P-11, as star spots that are occulted during the planet's transit cause detectable brightening in the transit light curve (Morris et al. 2017).

2. THE DATA

The data is collected by the *Kepler* mission and we extracted it using `lightkurve`, a Python package for Kepler and TESS data analysis (Lightkurve Collaboration et al. 2018).

3. HIERARCHICAL BAYESIAN MODEL

3.1. *The Model*

In this section, we will describe our Gaussian Process (GP) model and the likelihood calculation process used to estimate the model parameters.

We solve for a large set of parameters that includes the GP hyperparameters, the star's and the planet's orbital

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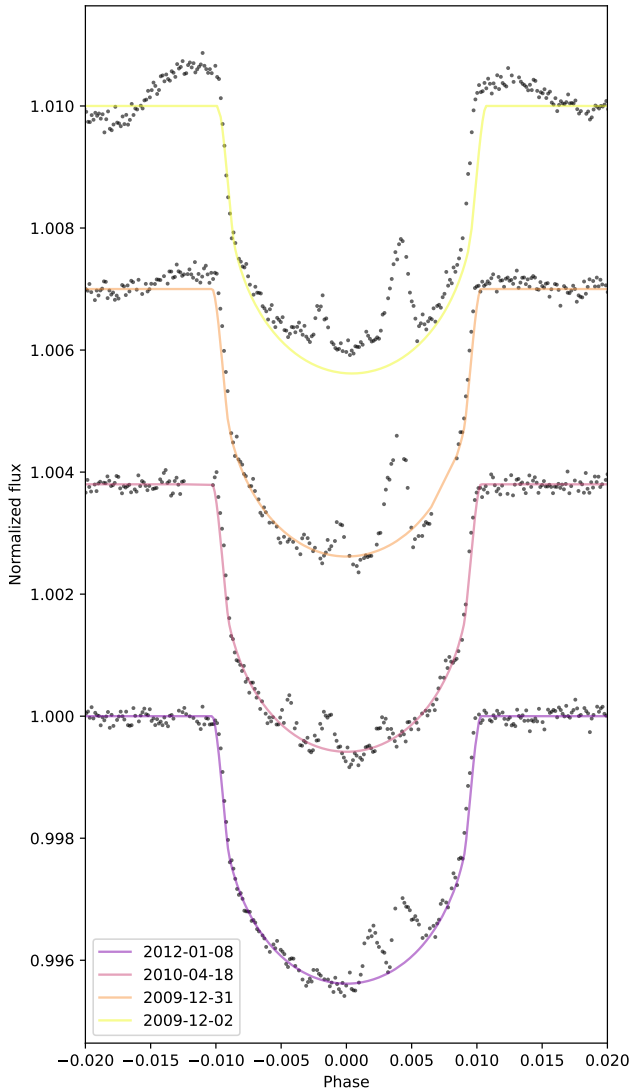

Figure 1. Transit fits with starry – no GPs.

parameters, respectively.

$$\boldsymbol{\Theta} = (\theta_\bullet \;\; \theta_\star \;\; \theta_p)^\top \quad , \tag{1}$$

Separately, these parameters are defined as

$$\boldsymbol{\theta_\star} = (i_\star \;\; m_\star \;\; u_1 \;\; u_2 \;\; P_\star)^\top \quad , \tag{2}$$

where $i_\star$ is the star's orbital inclination, $m_\star$ is the stellar mass in the units of the solar mass, u_1 and u_2 are limb-darkening coefficients, and $P_\star$ is the rotational period of the star.

$$\boldsymbol{\theta_p} = (i_p \;\; e \;\; \psi \;\; \omega \;\; P \;\; t_0 \;\; R_p/R_\star)^\top \quad , \tag{3}$$

where i_p is the planet's orbital inclination, e is its eccenticity, ψ is the stellar obliquity, ω is the argument of pericenter of the planet, P is the rotational period of the planet, t_0 is the transit start time, and $R_p/R_\star$ is the planet to star radius ratio.

We represent the GP hyperparameters as *physically interesting* set of parameters $\boldsymbol{\theta_\bullet}$ (Luger et al. 2021a):

$$\boldsymbol{\theta_\bullet} = (\mathbb{n} \;\; \mathbb{c} \;\; \mu_\phi \;\; \sigma_\phi \;\; \mathbb{r})^\top \quad , \tag{4}$$

where $\mathbb{n}$ is the number of starspots, $\mathbb{c}$ is their contrast (defined as the intensity difference between the spot and the background intensity, as a fraction of the background intensity), μ_ϕ and σ_ϕ are the mode and standard deviation of the spot latitude distribution, respectively, and $\mathbb{r}$ is the radius of the spots.

We assume that the prior over $\mathbb{f}_{true}$ follows a multivariate Gaussian distribution, with a mean vector of zeros and a covariance matrix $\boldsymbol{\Sigma}$. We use the quasi-periodic kernel to define the covariance matrix $\boldsymbol{\Sigma}$, which is defined by `StarryProcess`. We assume that the observations $\mathbb{f}_{obs}$ are corrupted by additive Gaussian noise, such that:

$$\mathbb{f}_{obs} = \mathbb{f}_{true} + \epsilon \tag{5}$$

where $\epsilon \sim \mathcal{N}(0, \sigma_n^2)$ is the noise term. Given the GP prior and the likelihood function, we can calculate the joint posterior distribution over the hyperparameters Θ and the true function $\mathbb{f}_{true}$ given the observed data $\mathbb{f}_{obs}$:

$$p(\Theta, \mathbb{f}_{true} \mid \mathbb{f}_{obs}) \propto p(\Theta) p(\mathbb{f}_{true} \mid \Theta) p(\mathbb{f}_{obs} \mid \mathbb{f}_{true}) \tag{6}$$

where $P(\Theta)$ is the prior distribution over the hyperparameters, $P(\mathbb{f}_{true} \mid \Theta)$ is the likelihood of the true function given the hyperparameters, and $P(f_{obs} \mid f_{true})$ is the likelihood of the observed data given the true function. We calculate the log-likelihood function, given by:

$$\ln p(\mathbb{f}_{obs} \mid \boldsymbol{\Theta}) = -\frac{1}{2}(\mathbb{f}_{obs} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbb{f}_{obs} - \boldsymbol{\mu}) - \frac{1}{2}\ln|\boldsymbol{\Sigma}| - \frac{n}{2}\ln(2\pi) \quad , \tag{7}$$

or, as defined in eq. 14 of (Luger et al. 2021a):

$$\ln \mathcal{L}_m(\Theta) = -\frac{1}{2}\mathbf{r}_m^\top(\Theta)\left[\boldsymbol{\Sigma}(\Theta)\right]^{-1}\mathbf{r}_m(\Theta) - \frac{1}{2}\ln\left|\boldsymbol{\Sigma}(\Theta)\right| - \frac{K}{2}\ln(2\pi) \quad , \tag{8}$$

where

$$\mathbf{r}_m(\boldsymbol{\Theta}) \equiv \mathbf{f}_m - \boldsymbol{\mu}(\boldsymbol{\Theta}) \tag{9}$$

is defined as the residual vector, Σ is the full covariance, which is defined as

$$\boldsymbol{\Sigma}(\Theta) \equiv \boldsymbol{\Sigma_y} + \boldsymbol{\Sigma_d} \tag{10}$$

where $\boldsymbol{\Sigma_y}$ is the covariance of the distribution over spherical harmonic coefficient vectors $\mathbb{y}$, and $\boldsymbol{\Sigma_d}$ is the data covariance, which is a diagonal matrix whose entries are the squared uncertainty σ_m^2 corresponding to each data point in the light curve. $|\cdots|$ denotes the determinant, and K is the number of data points in each light curve.

For faster calculations we rewrite the likelihood function using *the matrix conversion lemma*, also known as Woodbury-Sherman-Morrison identity (see e.g. Hogg et al. (2020)).

4. EXPERIMENTS

In this section, we describe the experiments we did on synthetic light curves before going ahead and modeling the real data. The synthetic light curves were generated by initializing a planet and a star of similar parameters as HAT-P-11 b and HAT-P-11 have using `starry`. Then we draw `StarryProcess` samples from a prior to get spherical harmonic coefficients and consequently the simulated flux.

4.1. *The issue of the normalization of light curves and units*

Before going on discussing the experiments produced for this paper, we first need to remind the reader of a subtlety of the flux normalization when we are given the raw light curves from telescopes. The problem and the ways to tackle it were described in Luger et al. (2021b) and Luger et al. (2021a).

Briefly, the problem of normalization of light curves is related to the fact that the observed flux from a star can vary due to a variety of factors such as atmospheric effects, instrumental noise, and changes in the intrinsic brightness of the star itself. These variations can make it difficult to compare light curves of different stars or even the same star observed at different times. To address this problem, astronomers typically normalize light curves by dividing the observed flux by some factor (the median or mean of the flux) that is assumed to be constant over time. However, as was described in Luger et al. (2021b), if a star has a single large equatorial spot of contrast c viewed at some high inclination (Luger et al. (2021b) used the value of 60°), and another star with a spot at the same location but with half the contrast *and* a large polar spot of comparable contrast, then the light curves for both of the stars become indistiguishable in the relative units astronomers observe them. In addition to normalizing the flux, Luger et al. (2021b) also added a *baseline* (1 in their case), which is the flux one would have gotten from a spotless star, and is an additive component to the Y_m^l.

In our experiments, we noticed that dividing by a constant factor *and* adding an additive baseline messes up the units due to the intricate transition between the spherical harmonics and flux unit bases. To address this issue, we opt for an alternative approach, wherein we impose a constraint upon the constant terms within the map, which correspond to a featureless star. Specifically, we achieve this by setting the prior distribution over these constant terms to be entirely uniform, thereby setting the first row of the precision matrix to zero.

4.2. *Short light curve*

For our initial experiment with minimal complexity, we created a synthetic light curve that involved a single planetary transit (therefore, *a short light curve*). Here, we examined two distinct sampling techniques: No-U-Turn Sampling (NUTS) using `pymc3` and Markov Chain Monte Carlo (MCMC) using `emcee`. The MCMC technique exhibited a faster convergence rate of the walkers, despite the fact that it still required a significant amount of time.

4.3. *Long light curves (multiple transits)*

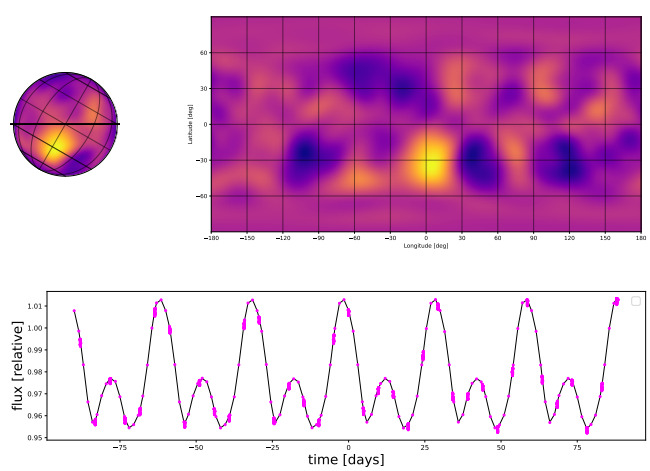

Figure 2. A synthetic dataset.

5. RESULTS AND DISCUSSION

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Table 1. The free parameters for the synthetic data, their true values, and their priors.

Parameter	True value	Prior distribution
i_p	90°	Planetary Inclination
e	0	$\sim \mathcal{U}(0, 0.4)$
P	4.887802443	Period
t_0	0.0	$\sim \mathcal{U}(-0.2, 0.2)$
R_p	0.02732	Logarithmic
m_p	0.81×10^{-3}	$\sim \mathcal{U}(10^{-4}, 10^{-3})$
$i_\star$	90°	Stellar Angle
λ	30°	Stellar Angle
$P_\star$	30	Period
$\mathbb{r}$	20°	$\sim \mathcal{U}(0.1, 1)$
$\mathbb{c}$	0.5	$\sim \mathcal{U}(0.01, 1)$
$\mathbb{n}$	1	$\sim \mathcal{U}(0, 10)$
μ_ϕ	30	$\sim \mathcal{U}(0, 80)$
σ_ϕ	5	$\sim \mathcal{U}(1, 10)$

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